

PROBLEM 21.4. Let $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ be the set of integers. Do $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality? If so, give a bijection between them.

PROBLEM 21.5. Let $\mathbb{Q}$ be the set of rational numbers, i.e. numbers of the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$, where two such expressions are considered the same if they are equal as fractions. (So, for instance, $\frac{1}{2}$ and $\frac{2}{4}$ are the same rational number.) Do $\mathbb{N}$ and $\mathbb{Q}$ have the same cardinality? If so, give a bijection between them.

WORKSHEET 21

Basic Infinite Set Theory

We say that two sets A and B have the same *cardinality* if there is a way of matching up every element of A with a unique element of B , and vice versa. See Figure 1 for an example. We write $|A| = |B|$ if A and B have the same cardinality. The matching function giving us an element of B for each element of A is called a *bijection*.

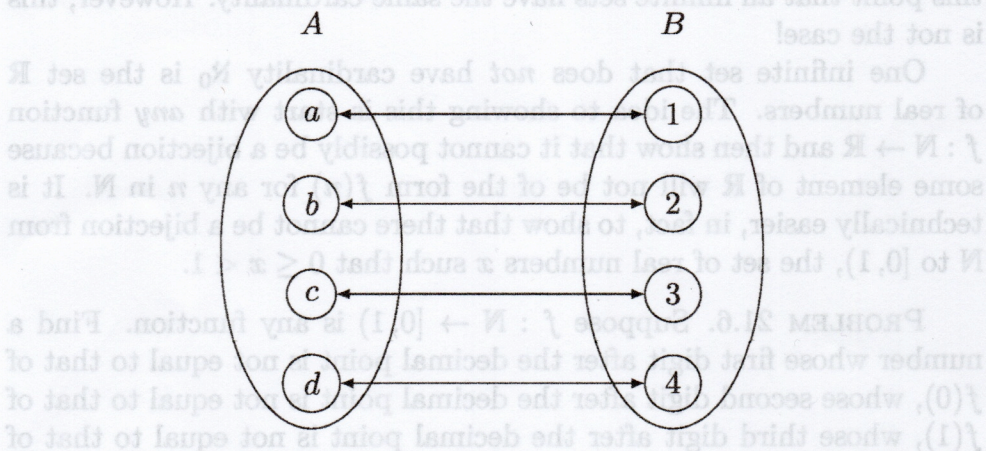


Figure 1. A one-to-one correspondence between two sets.

PROBLEM 21.1. Show that if A and B are finite sets, then $|A| = |B|$ if and only if A and B have the same number of elements.

PROBLEM 21.2. Show that if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.

If A and B are finite sets, then the concept of cardinality agrees with what you already know about counting. But when they are infinite, things get weirder!

PROBLEM 21.3. Let $\mathbb{N} = \{0, 1, 2, \dots\}$ be the set of nonnegative integers, and let $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ be the set of positive integers. Do $\mathbb{N}$ and $\mathbb{Z}^+$ have the same cardinality? If so, give a bijection between them.

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So far (spoiler alert!), all the infinite sets we've discussed have the same cardinality. We call this cardinality $\aleph_0$. The letter $\aleph$ is *aleph*, the first letter of the Hebrew alphabet. You may find it surprising at this point that, say, $\mathbb{N}$ and $\mathbb{Q}$ have the same cardinality, between $\mathbb{Q}$ seems to be much larger than $\mathbb{N}$, so you would be forgiven for suspecting at this point that all infinite sets have the same cardinality. However, this is not the case!

One infinite set that does *not* have cardinality $\aleph_0$ is the set $\mathbb{R}$ of real numbers. The idea to showing this is start with *any* function $f : \mathbb{N} \rightarrow \mathbb{R}$ and then show that it cannot possibly be a bijection because some element of $\mathbb{R}$ will not be of the form $f(n)$ for any n in $\mathbb{N}$. It is technically easier, in fact, to show that there cannot be a bijection from $\mathbb{N}$ to $[0, 1)$, the set of real numbers x such that $0 \leq x < 1$.

PROBLEM 21.6. Suppose $f : \mathbb{N} \rightarrow [0, 1)$ is any function. Find a number whose first digit after the decimal point is not equal to that of $f(0)$, whose second digit after the decimal point is not equal to that of $f(1)$, whose third digit after the decimal point is not equal to that of $f(2)$, and so on.

We would like to conclude that the resulting number is not of the form $f(n)$ for any n . (Why?) But there is a slight issue, namely that some numbers have more than one decimal expansion. For instance, $0.09999\dots = 0.1$. However, the only numbers with two decimal expansions are ones where one expansion ends with all 9's, and the other ends with all 0's.

PROBLEM 21.7. Modify the construction from the previous problem to avoid this issue.

PROBLEM 21.8. Conclude that $[0, 1)$ does not have the same cardinality as $\mathbb{N}$.

PROBLEM 21.9. Show that $[0, 1)$ has the same cardinality as $\mathbb{R}$.

PROBLEM 21.10. Conclude that $\mathbb{N}$ does not have the same cardinality as $\mathbb{R}$.

We call the cardinality of $\mathbb{R}$ *the continuum* and denote it either $\mathfrak{c}$ or $2^{\aleph_0}$.

Another interesting construction is the *power set*. Given any set A , its *power set* $\mathcal{P}(A)$ is the set of all subsets of A . For instance, if $A = \{1, 2\}$, then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Thus the cardinality of $\mathcal{P}(A)$ is 4. Note that $\emptyset = \{\}$ denotes the empty set, i.e. the set containing no elements.

PROBLEM 21.11. Suppose A is a finite set with n elements. Show that $\mathcal{P}(A)$ is a finite set. How many elements does it have?

PROBLEM 21.12. Show that if A is any set, then A and $\mathcal{P}(A)$ do not have the same cardinality. (You can use a very similar technique to the one we used to show that $\mathbb{N}$ and $[0, 1)$ do not have the same cardinality.)

PROBLEM 21.13. Show that the set of infinite binary sequences has the same cardinality as the set of infinite ternary (i.e. base 3) sequences.

PROBLEM 21.14. Show that $\mathbb{R}$ and $\mathcal{P}(\mathbb{N})$ have the same cardinality.

Basic Infinite Set Theory

Alexander Friesen

*Is this the correct term?
†I know, not a thing...

21.1: Assume the contrary:



Here, matching any element to the extra on the right requires a single element to have two matches, which is impossible. This is the same case for any other two non-cardinal* sets.

21.2: If $|A| = |B|$, and every set in B maps to a set in A , then it is possible to replace every term in set B with every corresponding term in set A . Because $|B| = |C|$, then replacing B with A validly gives that $|A| = |C|$.

21.3: Yes! Here is a graph of the matching functions:

N	0	1	2	3	4	5	6	7	8	9	10	...	$\infty - 1$ [†]
Z ⁺	1	2	3	4	5	6	7	8	9	10	11	...	∞

* For $Q \in [0, 1)$. Can work for infinite as long as $[0, 1)$ works.

21.4: Yes! Here is a graph again:

N	0	1	2	3	4	5	6	7	8	$\dots \infty$
Z	0	1	-1	2	-2	3	-3	4	-4	$\dots \pm \infty$

21.5: Again, here is a representative graph:

N	0	1	2	3	4	5	6	7	8	$\dots \infty$
Q	0	$\frac{1}{1}$	$\frac{-1}{1}$	$\frac{1}{2}$	$\frac{-1}{2}$	$\frac{1}{3}$	$\frac{-1}{3}$	$\frac{2}{3}$	$\frac{-2}{3}$	$\dots \pm \infty$

So far, all of the sets covered are countable infinities, or $\aleph_0$.

21.6: Randomly assign $f(n)$'s:

$$f(0) = 0.123456789101112\dots$$

$$f(1) = 0.23578129993221\dots$$

$$f(2) = 0.357891244345554\dots$$

$$f(3) = 0.\text{string of numbers}\dots$$

$$f(n) = 0.\text{string of numbers}\dots$$

$$\text{Not in } f = 0.248ABCDEFGHIJ\dots$$

where a letter is not equal to the $(n+1)$ th digit of $f(n)$.

21.7: The number does not include

a long enough string of 9's

to change its value to

one in f , so we are safe.

21.8: Because the number created is not in $\mathbb{N}$ but is in $[0, 1)$, it cannot be corresponding to a number in $\mathbb{N}$. Therefore, $[0, 1)$ and $\mathbb{N}$ don't have the same cardinality (set $[0, 1) > \mathbb{N}$.)

21.9: If $[0, 1) \neq \mathbb{N}$, then it $\neq \aleph_0$. $\mathbb{R} \neq \aleph_0$, so both $[0, 1)$ and $\mathbb{R}$ are greater than $\aleph_0$ (lest they be finite.)

To be greater than countable ∞ , both $[0, 1)$ and $\mathbb{R}$ have to be uncountable infinity, or $\mathcal{C}$.

Therefore, because $\mathcal{C} = \mathcal{C}$, $[0, 1) = \mathbb{R}$.

21.10: If $[0, 1) \neq \mathbb{N}$ and $[0, 1) = \mathbb{R}$, then by definition $\mathbb{N} \neq \mathbb{R}$.

21.11: Each power set $P(A)$ turns out to always have 2^A subsets, because each subset can either "choose" to include or leave out any term in A , leading to $2 \times 2 \times 2 \dots = 2^A$ choices. 2^A is finite if A is, so $P(A)$ is finite.

21.12: The power set will always include the "empty set", $\emptyset$, which counts as a term itself. However, $\emptyset$ is never a term in a regular set, so the power set will always be greater by (at least) 1 term. Thus, $A \neq P(A)$.

21.13: Any particular binary or ternary sequence can map to a base 10 number:

Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011
Ternary	0	1	2	10	11	12	20	21	22	100	101	102
Base 10 Equivalent	0	1	2	3	4	5	6	7	8	9	10	11

Because $B_2 = B_{10}$ and $B_3 = B_{10}$, $B_2 = B_3$ and the two bases of $\mathbb{N}$ have the same cardinality.

21.14: Each set in $P(\mathbb{N})$ can be represented as an ∞ binary string where a 0 in position n means n is not in a subset in $P(\mathbb{N})$, and a 1 in position n means vice versa.

*This requires binary decimals, such as $0.010101... = \frac{1}{3}$.
 Not sure on the set formation algorithm yet!
 21.14(cont.): For example, $\{2, 4, 5\}$
 # Irrational, not imaginary.

$= 0101100000000...$

Now, we have to show that every one of these binary strings represents an $n \in \mathbb{R}$.

Of course, any string with trailing zeroes or a repetitive pattern is part of $\mathbb{Q}$, the first being integers/decimals and the second being rational repetitive decimals.* Irrational

numbers are infinite (with no set of trailing zeroes) and without pattern, in binary.

(Note: a guess on the binary decimal formation is to start after the decimal point and read. Every 0, take the lower $\frac{1}{2}$ and every 1 the upper $\frac{1}{2}$ of the range, starting with $n - n+1$

before the decimal point.) $\mathbb{R} = \{\mathbb{N}, \mathbb{Q}, \mathbb{Z}, \mathbb{I}^+\} = P(\mathbb{N})$